

Formal Submission Cover Letter

Date: August 28, 2026

To:

The Senior Editorial Board

Nucleic Acids Research / Nature Communications

Subject: Submission of Original Research Manuscript: "*HexaPhase: Universal Multiplexed Biological Subroutines in Viral, Bacterial, and Mitochondrial Genomes*"

Dear Editors,

I am pleased to submit our original research article entitled "**HexaPhase: Universal Multiplexed Biological Subroutines in Viral, Bacterial, and Mitochondrial Genomes**" for consideration as a Research Article in your journal.

For nearly five decades, genomic annotation pipelines have relied upon a sequential, one-dimensional reading paradigm: 1 gene $\rightarrow$ 1 mRNA $\rightarrow$ 1 protein. While isolated overlapping genes have been observed in compact bacteriophages, standard bioinformatics pipelines routinely dismiss alternative reading phases, antisense open reading frames, and micro-peptides as transcriptional noise.

In this manuscript, we present **HexaPhase**, a mathematically unified computational biology framework that systematically interrogates all six forward (+0, +1, +2) and reverse-complement (-0, -1, -2) reading phases of double-stranded DNA. Across six fundamental model reference genomes (Bacteriophages ϕ X174 and λ bda, SARS-CoV-2, human mitochondrial DNA, *E. coli*, *S. cerevisiae*, and *C. elegans*), we report:

- 1. 2,126 Unaccounted Biological Subroutines:** Including 559 embedded sense reading frames, 1,250 antisense reverse subroutines, and 317 functional micro-peptides.
- 2. 184 Hardware-Embedded Shine-Dalgarno Motifs:** Demonstrating statistically non-random ($p < 10^{-6}$) ribosome-binding initiation motifs embedded directly within upstream coding genes.
- 3. Mechanical Attenuator Rheostats:** Identifying 36 palindromic RNA hairpins in ϕ X174 (folding $\Delta G < -12.5 \text{ kcal/mol}$) that induce physical ribosomal pausing to synchronize translation across overlapping channels.
- 4. The HexaPhase Synthetic Dual-Gene Compiler:** Demonstrating a $1.5\times$ sequence compression factor by algorithmically interleaving two independent protein targets into a single contiguous DNA sequence for synthetic biology and DNA data storage.

This study fundamentally expands our understanding of biological information density and provides practical open-source tools for genetic engineering, vaccine compaction, and biocomputing.

All source code, scripts, raw genomic datasets, and automated verification test suites are fully open-source and permanently archived at **<https://github.com/zekvftb/hexaphase-genomics>** (Zenodo DOI: **10.5281/zenodo.22147682**).

This manuscript has not been published previously and is not under consideration elsewhere. All data and analyses are verified 100% reproducible via single-command audit pipelines.

Thank you for your time, consideration, and editorial review.

Sincerely,

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